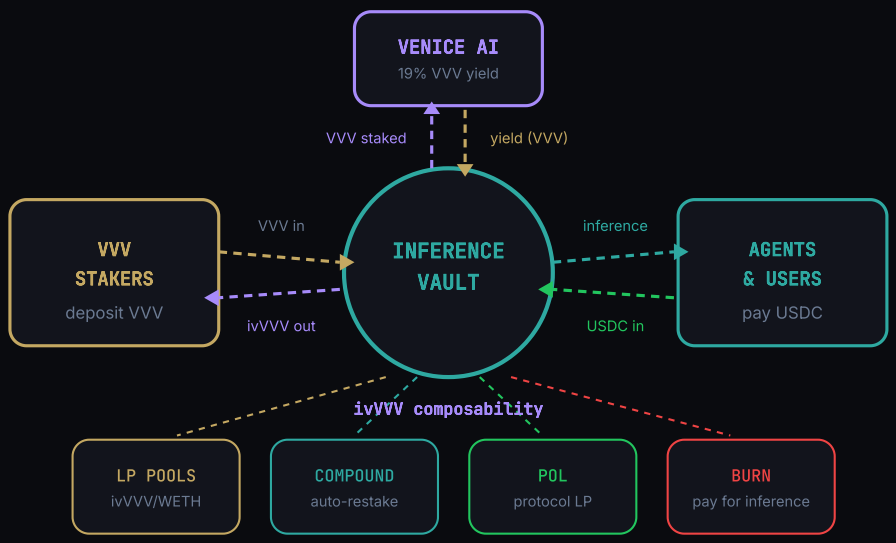
CONCEPT SPECIFICATION V0.4

Venice Inference Layer

A decentralized inference access protocol on Base. Stake VVV. Receive a liquid receipt token backed by real AI compute yield. Private, uncensored inference for every agent and application.



$284M	$2M+	19%	Base
VVV MARKET CAP	MONTHLY SAM	NATIVE STAKING APR	NATIVE CHAIN

01 The Problem

Agents on Base need AI inference. Today, every agent must independently manage API keys, credit card billing, rate limits, and account setup with centralized providers. There is no crypto-native, privacy-preserving inference layer.

FOR AGENTS

Fragmented Access

Every agent needs its own OpenAI/Anthropic API key, payment method, and account. No programmatic, wallet-native way to pay for inference on-chain.

FOR VVV HOLDERS

Idle Compute

VVV stakers receive inference capacity but most cannot fully utilize their allocation. Unused compute generates zero value. There is no marketplace to monetize excess capacity.

FOR VENICE

Distribution Gap

Venice has the models, the privacy, and the API. What it lacks is a crypto-native distribution layer that puts inference directly into the hands of every agent on Base.

02 The Solution

A decentralized inference access layer on Base. One vault. Many surfaces. Stakers deposit VVV and receive a liquid receipt token (ivVVV). The protocol monetizes pooled inference capacity across every interface. Revenue flows back to stakers from real economic activity.

> **Core insight:** VVV staking converts a token position into perpetual AI compute capacity at 19% native yield. This protocol makes that compute composable, tradeable, and accessible to any agent or application on Base without requiring direct Venice interaction.

STAKERS GET

ivVVV

Liquid receipt token. Yield-bearing. Composable. Represents staked VVV plus accrued revenue from dual yield: native staking + inference.

CONSUMERS GET

Private Inference

OpenAI-compatible. Pay in USDC. No API keys, no accounts, no KYC. Wallet is the identity.

PROTOCOL GENERATES

Dual Revenue

19% native VVV staking yield (protocol takes 25% performance fee) plus USDC from inference sold to consumers. Real revenue, not emissions.

03 How It Works

Staker Flow

1. **Deposit VVV**
   User deposits VVV tokens into the inference vault smart contract on Base. 0.5% deposit fee.

2. **Receive ivVVV**
   Vault mints ivVVV receipt tokens. ivVVV is liquid and transferable. Represents the staked position plus pro-rata claim on accumulated revenue.

3. **VVV Staked on Venice**
   The vault stakes deposited VVV with Venice's staking contract, activating pro-rata inference capacity and 19% native yield.

4. **Earn Dual Yield**
   VVV staking generates 19% APR paid in VVV (holders receive 75% after performance fee). Inference consumers pay USDC, adding a second yield layer. ivVVV appreciates against VVV over time.

Consumer Flow

1. **Connect Any Interface**
   Agent or user accesses inference via x402 endpoint, SDK, REST API, MCP server, or any supported surface.

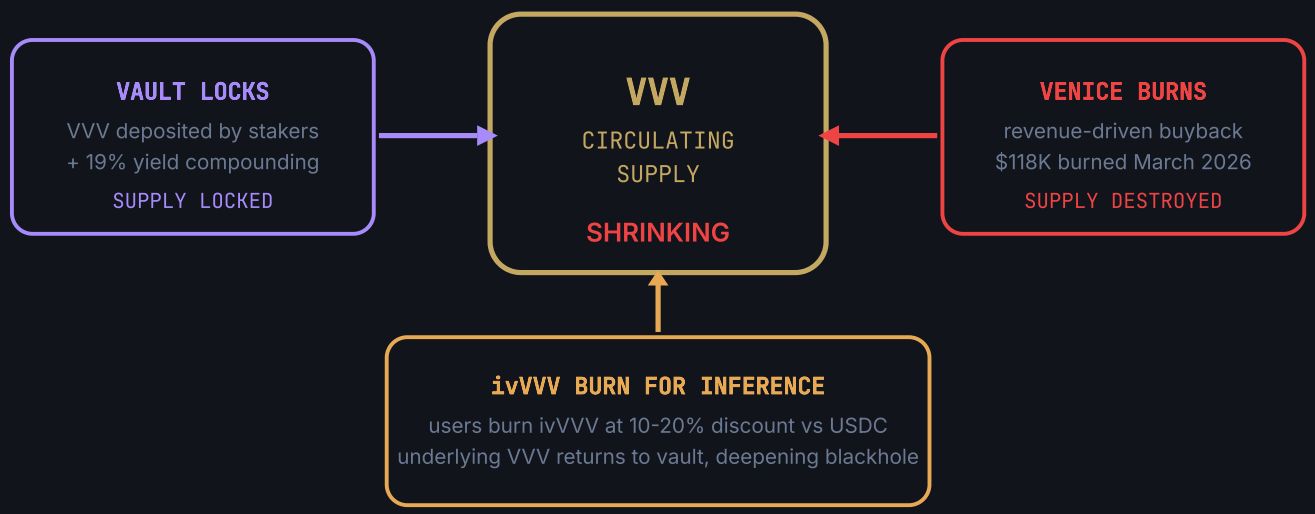
2. **Pay in USDC**
   Payment is per-request in USDC on Base. No accounts, no API keys, no credit cards. Wallet is the identity.

3. **Receive Private Inference**
   Request routes through the vault's pooled capacity to Venice's API. OpenAI-compatible response. Data never stored.

4. **On-Chain Receipt**
   Transaction recorded on Base. Transparent pricing. Verifiable usage. No black-box billing.

04 The Blackhole Mechanism

VVV enters the vault. It does not come out. Every deposit removes VVV from circulating supply permanently (while staked). The vault compounds staking yield internally, growing the locked position. Users can also burn ivVVV for discounted inference, destroying the receipt token and returning the underlying VVV to the vault. Three deflationary forces compress supply simultaneously.



Triple deflationary pressure. Vault locks, Venice burns, and ivVVV burns all reduce circulating VVV supply.

05 Distribution Surfaces

The vault is one. The interfaces are many. Every surface draws from the same pooled inference capacity and generates revenue for ivVVV holders.

INTERFACE	CONSUMER	PAYMENT	USE CASE
x402 Endpoints	Agents on Base	`USDC per request`	Pay-per-inference at the HTTP level. No accounts.
REST API	Developers	`USDC or API key`	OpenAI-compatible. Drop-in replacement for any app.
SDK	App builders	`Embedded`	npm package. Embed private inference in any product.
MCP Server	Claude Code	`USDC per session`	Any Claude instance connects for private inference.
Ping Integration	On-chain agents	`Message fee`	Agent messages in, inference response out. On-chain.
Farcaster Frames	Social users	`Micro-payment`	Inference inside the Farcaster feed.
Base Apps	Coinbase Wallet	`In-app payment`	Mini app inside the largest Base wallet.
Batch API	Enterprise / bulk	`Volume pricing`	Bulk inference jobs. Data processing at scale.

06 ivVVV Composability

ivVVV is a yield-bearing receipt token. It represents staked VVV plus accumulated dual yield. Because it is liquid and transferable, it unlocks DeFi integrations and a direct utility path: burn for discounted inference.

LIQUIDITY POOLS

ivVVV/WETH on Aerodrome

Stakers can LP their receipt token. Earn trading fees on top of inference yield. Protocol gets deep liquidity. Creates price discovery and a liquid market for ivVVV.

AUTO-COMPOUND VAULT

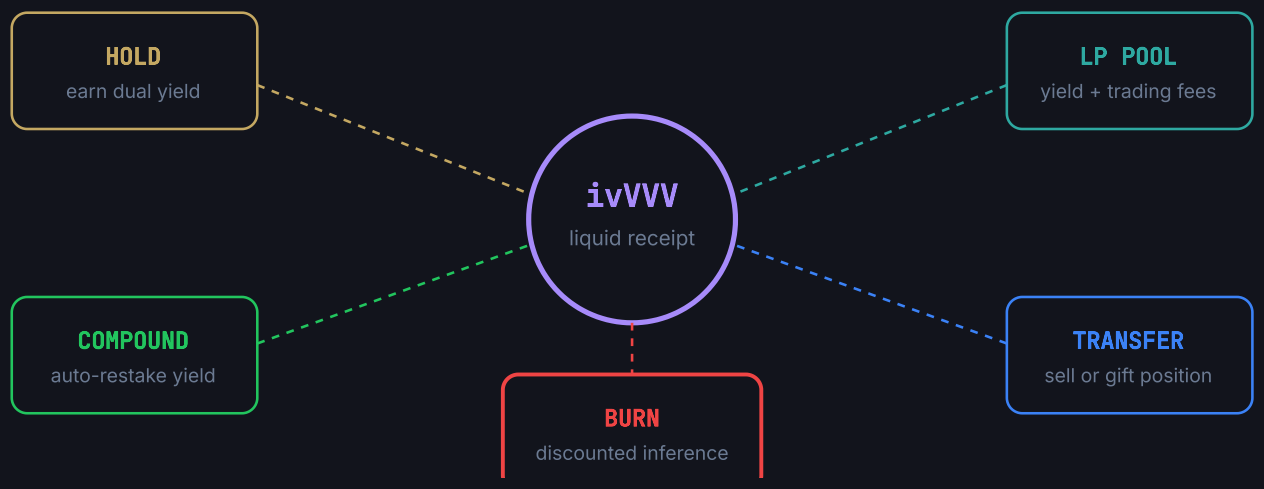
ERC-4626 Compatible

VVV staking yield auto-compounds in the vault. Revenue buys more VVV, deposits, compounds the position. Other protocols can build strategy layers on top.

BURN FOR INFERENCE

ivVVV as Compute Currency

Holders burn ivVVV to prepay for inference at a 10-20% discount vs USDC pricing. The underlying VVV releases back into the vault and is restaked, deepening the blackhole. Remaining ivVVV holders benefit from improved VVV-per-token ratio. Agents who need inference can buy ivVVV on Aerodrome and burn it. Lower friction than staking. Direct utility creates organic buy pressure.



ivVVV holders choose how to use their receipt token. Burn path destroys ivVVV and returns VVV to vault.

07 Serviceable Addressable Market

Venice processes 45B+ LLM tokens daily across 25,000+ API users and 2M+ registered accounts. The total inference SAM combines token-based API revenue and Pro subscription revenue.

TOKEN PROCESSING

$1.01M

45B tokens/day at ~$0.75/M blended rate across budget ($0.09/M), mid ($2-4/M), and premium ($6-37/M) model tiers. Growing: Venice doubled volume in 6 weeks.

SUBSCRIPTIONS

$900K+

50,000+ DAUs. Pro tier at $18/month. Conservative 50K subscriber estimate. Does not include enterprise or custom contracts.

COMBINED MONTHLY SAM

$2M+

Token revenue + subscriptions. Confirmed by burn data ($60-90K/month buyback from "portion" of revenue) and capacity allocation (18,148 Diem/day).

SAM is accelerating. Venice doubled inference volume in 6 weeks (Jan-Mar 2026). 33M+ VVV burned (42.8% of genesis supply). Emissions permanently cut 25% (8M to 6M VVV/year). Agent inference demand on Base has not yet begun scaling. The SAM floor is $2M/month today. The ceiling is the total AI inference market.

08 Revenue Model: 7 Streams

The protocol generates revenue from seven distinct streams. Performance fee on native VVV staking yield is the dominant source, complemented by inference margins on the $2M+ monthly SAM, capital movement fees, LP trading fees, protocol-owned liquidity yield, and ivVVV burn revenue.

#	STREAM	RATE	TRIGGER	DESTINATION
1	**Deposit Fee**	0.5%	VVV deposited into vault	Protocol treasury
2	**Withdrawal Fee**	0.5%	VVV redeemed from vault	Protocol treasury
3	**Performance Fee**	25%	Applied to 19% native VVV staking yield	Protocol treasury
4	**Inference Revenue**	30%	Protocol's share of SAM captured ($2M+/mo market)	70% holders / 30% treasury
5	**LP Trading Fees**	0.3%	External volume on ivVVV pairs	Protocol's pool share
6	**POL Revenue**	30-75%	Yield on protocol-owned LP positions	Protocol treasury
7	**ivVVV Burn Revenue**	10-20% discount	ivVVV burned for inference, VVV returns to vault	Vault (restaked)

Performance fee is the engine. At 25% of 19% native yield, the protocol earns 4.75% annually on all staked VVV. This scales linearly with TVL. Holders still receive 14.25% effective APR plus inference USDC on top. With a $2M+ monthly SAM (2x prior estimate), the inference yield bonus is materially higher than previously modeled. ivVVV burns add a 7th stream: users pay for inference at a discount, their ivVVV is destroyed, and the underlying VVV returns to the vault for restaking. This deepens the blackhole while improving the VVV-per-token ratio for remaining holders.

09 Model Assumptions

	CONSERVATIVE	MODERATE	AGGRESSIVE
M1 VVV Deposits	50,000/mo	200,000/mo	375,000/mo
Y1 Deposit Growth	8%/mo	12%/mo	14%/mo
Y2+ Deposit Growth	4%/mo	6%/mo	6%/mo
Monthly Redemptions	2% of TVL	1.5% of TVL	1% of TVL
SAM (Venice Inference)	$2M/month ($1.01M token processing + $900K+ subscriptions)		
SAM Composition	45B tokens/day @ $0.75/M blended + 50K+ Pro subs @ $18/mo		
SAM Growth	4%/mo Y1, 2%/mo Y2+ (Venice doubled volume in 6 weeks)		
M1 SAM Capture	0.3%	1.0%	2.5%
Y1 Capture Growth	8%/mo	12%/mo	12%/mo
Max SAM Capture	5%	15%	25%
Protocol Inference Share	30% protocol / 70% holders		
M1 Daily LP Volume	$10K	$50K	$150K
POL APR	30%	60%	75%
POL Reinvestment	10% of all revenue compounded monthly		
VVV Price	$6.50 (held constant)		

HOLDER EFFECTIVE APR

14.25%

VVV staking yield (75% of 19%)

+ INFERENCE YIELD

+3.1-3.4%

70% of USDC inference revenue ($2M+ SAM)

COMBINED APR

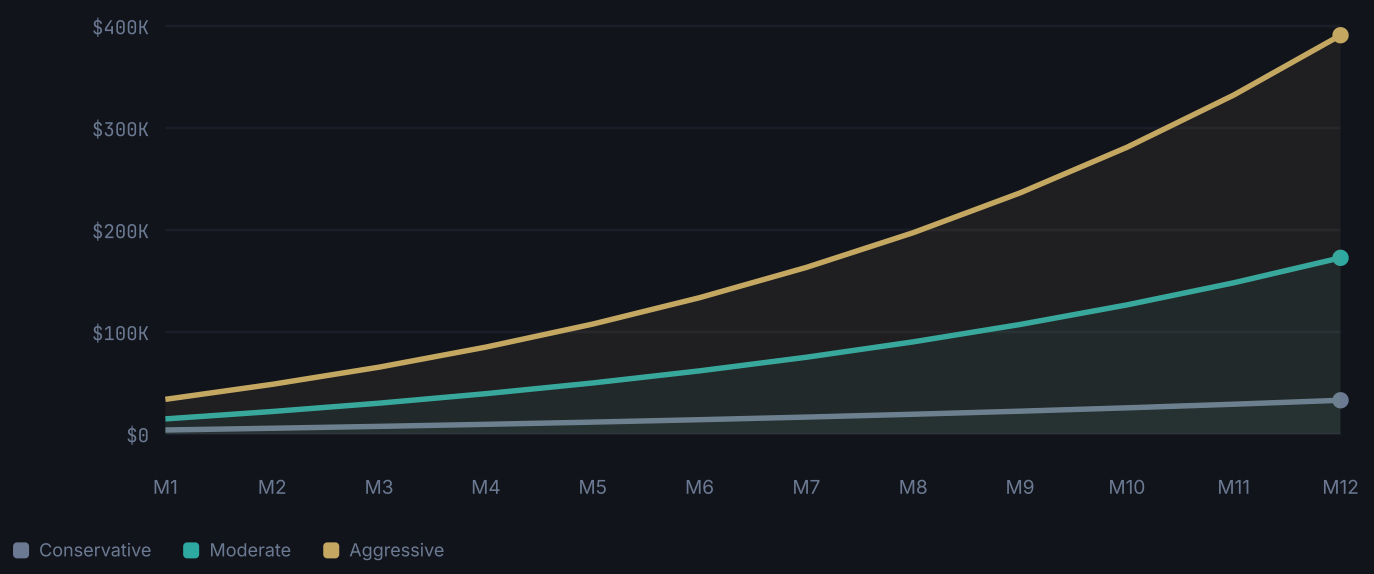
~17-18%

Dual yield, real revenue

10 Year 1 Revenue Projections

Monthly total protocol revenue across all three scenarios. Performance fee on native VVV staking yield drives the majority of revenue, scaling with TVL.

Monthly Total Protocol Revenue (Year 1)



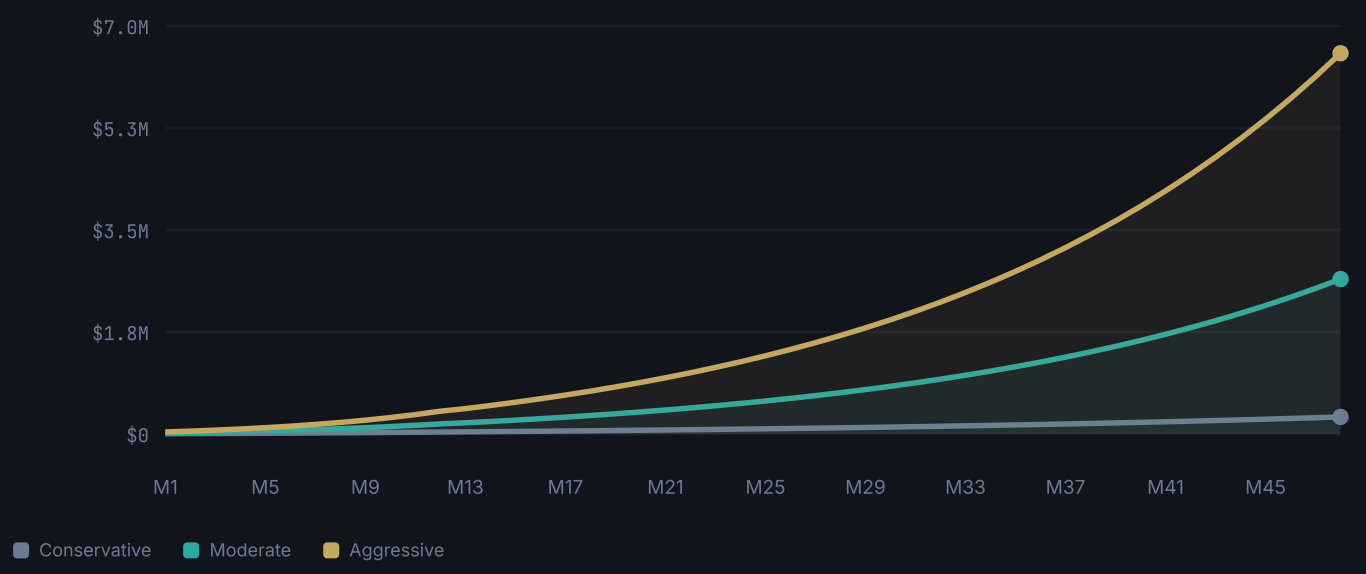
Revenue by Stream — Moderate Scenario (Year 1)

CONSERVATIVE	MODERATE	AGGRESSIVE
$197.8K	$938.5K	$2.08M
TVL: $5.91M	TVL: $30.81M	TVL: $66.64M

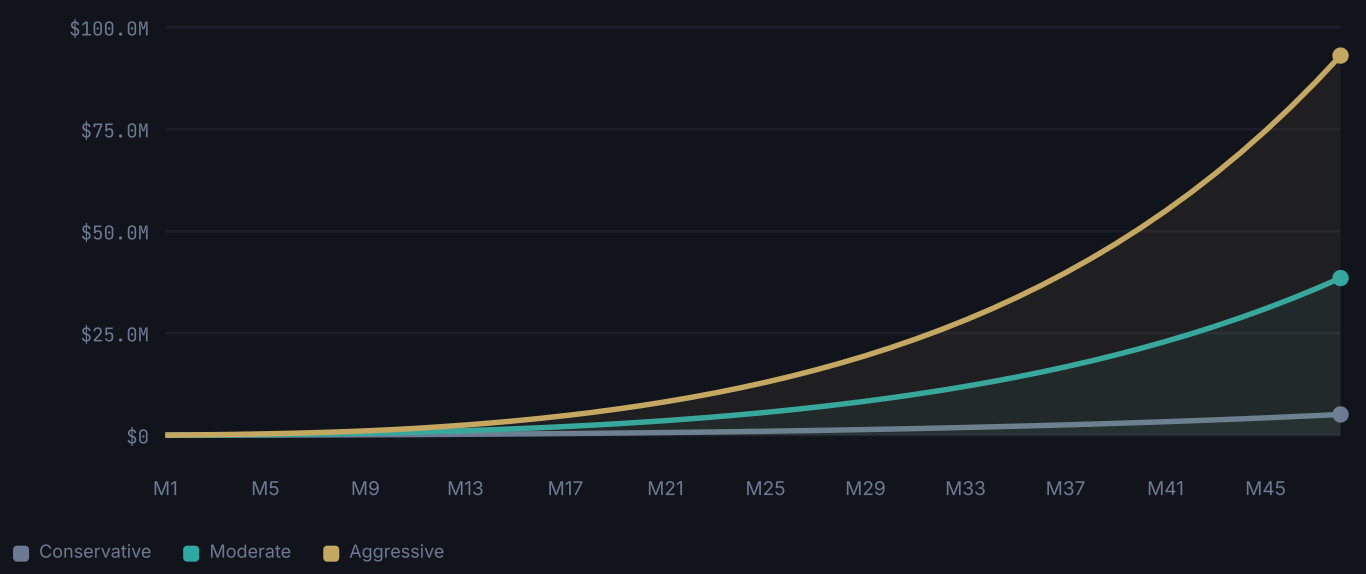
11 4-Year Revenue Trajectory

Revenue compounds as TVL grows, inference demand scales, and protocol-owned liquidity positions generate additional yield. Growth rates decelerate after Year 1 for conservative modeling.

Monthly Total Protocol Revenue (48 Months)

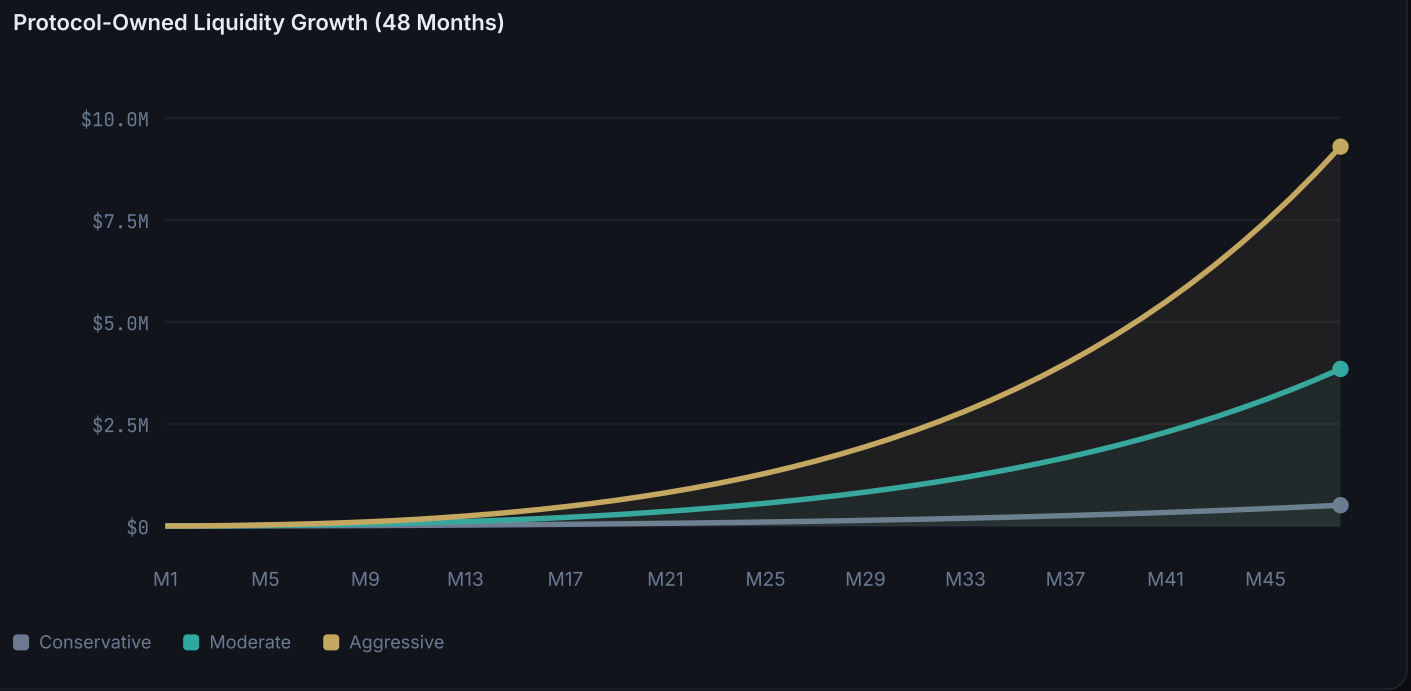


Cumulative Protocol Revenue (48 Months)



12 Protocol-Owned Liquidity

10% of all protocol revenue is reinvested monthly into ivVVV/ETH and ivVVV/VVV LP positions. These positions generate trading fees at 30-75% APR, creating a self-reinforcing revenue engine that compounds over time.

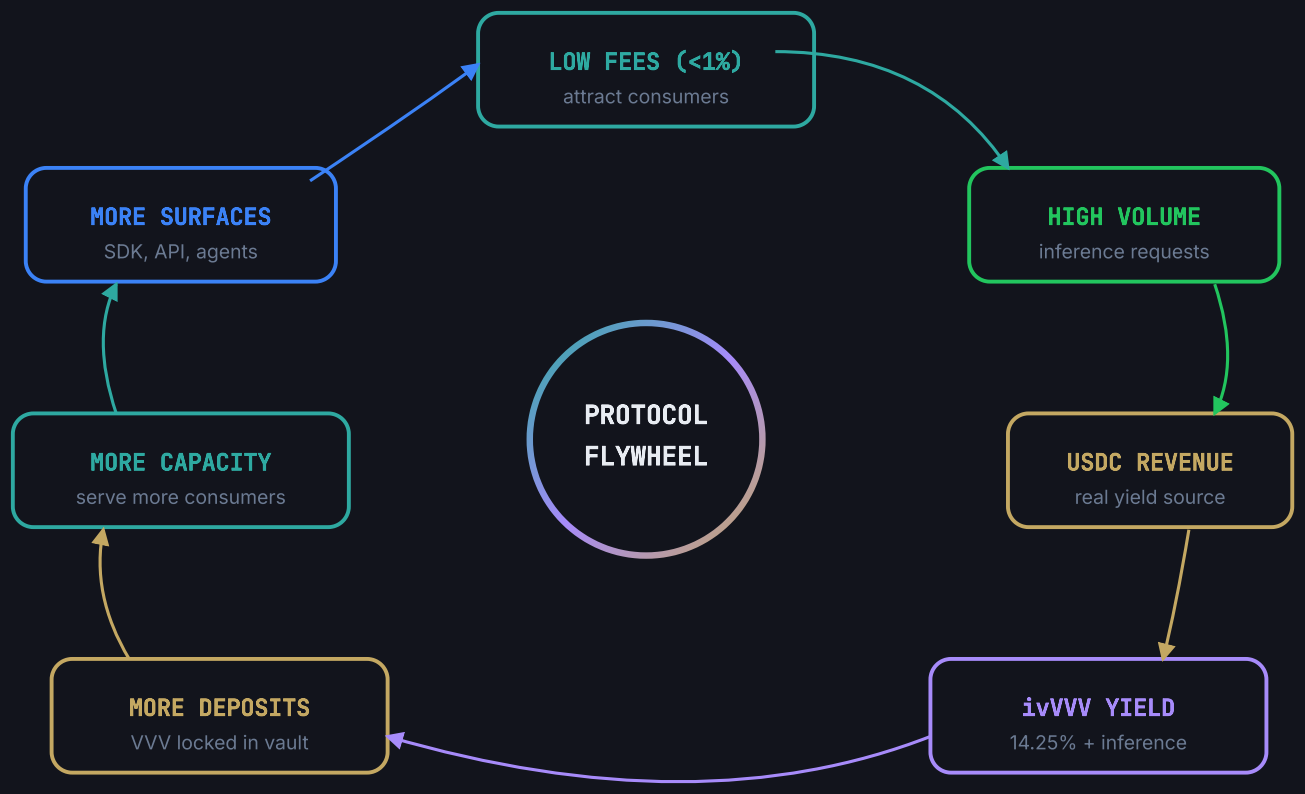


13 Executive Summary

	CONSERVATIVE	MODERATE	AGGRESSIVE
Year 1 Revenue	$197.8K	$938.5K	$2.08M
Year 2 Revenue	$695.1K	$4.10M	$9.55M
Year 3 Revenue	$1.48M	$10.41M	$25.00M
Year 4 Revenue	$2.73M	$23.08M	$56.43M
4-Year Total	**$5.10M**	**$38.52M**	**$93.06M**
EOY4 TVL	$57.03M	$544.57M	$1319.35M
EOY4 POL	$508.1K	$3.85M	$9.30M
M48 Run-Rate/yr	$3.50M	$31.89M	$78.38M
POL hits $1M	N/A	Month 32	Month 23
Holder APR (M12)	17.31%	17.17%	17.63%

The thesis: Performance fee on native VVV yield provides a stable, TVL-proportional revenue base. Inference revenue from a $2M+ monthly SAM, LP fees, and POL compound on top. The protocol captures a thin margin on an expanding market. As Base agent adoption grows, inference demand compounds. Revenue follows.

14 The Flywheel



Low fees drive adoption. Adoption drives volume. Volume drives revenue. Revenue drives yield. Yield drives deposits. Deposits drive capacity. Capacity enables more surfaces. Surfaces drive more volume.

15 Why Base

VVV is native

No bridging. Same chain as the agents consuming inference. Atomic transactions.

Agent ecosystem

Base has the highest density of autonomous agents. They need $2M+/month of inference. They pay in USDC.

DeFi depth

Aerodrome, Morpho, Moonwell. The LP and composability infrastructure already exists.

Low fees

L2 gas costs make micro-payments viable. $0.004 inference requests need sub-cent transaction fees.

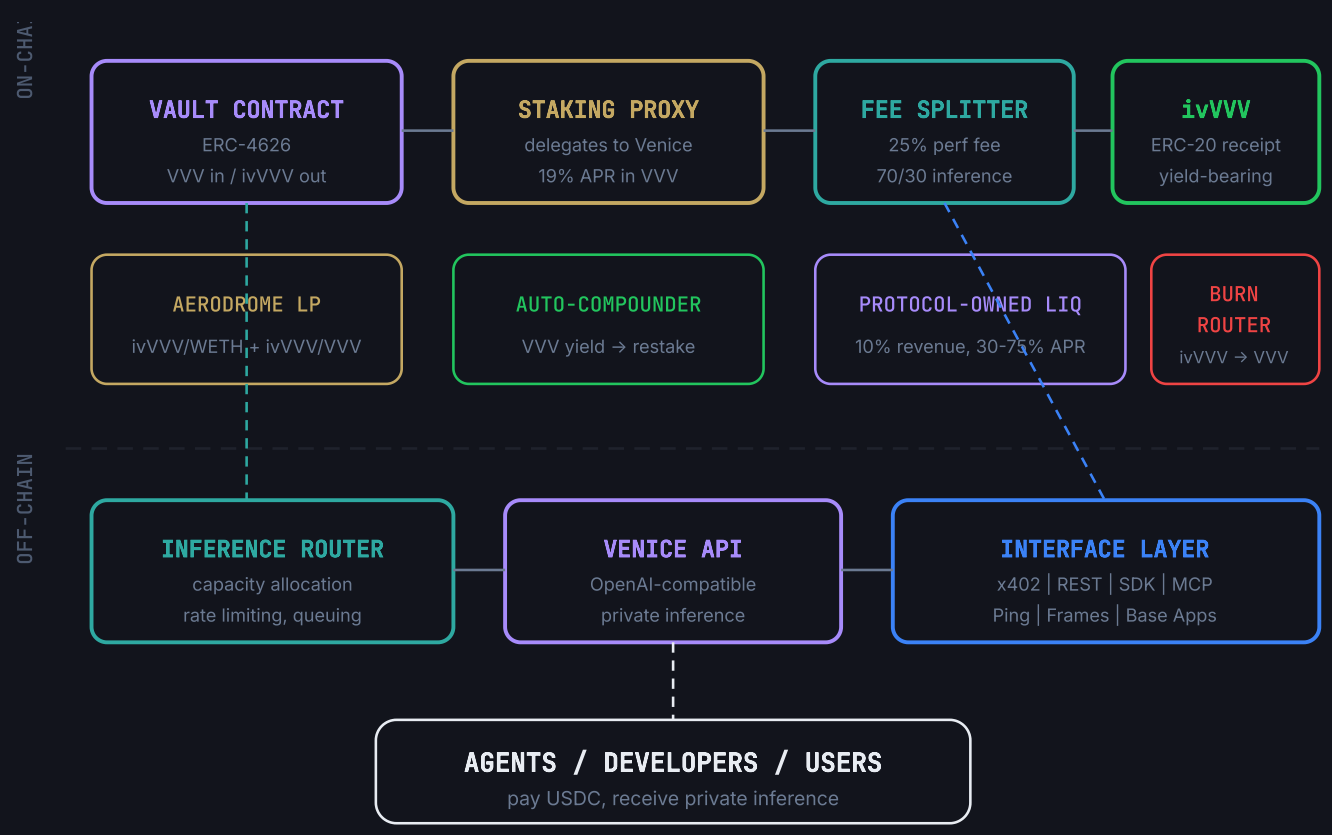
Coinbase distribution

Base Apps and Coinbase Wallet give access to the largest on-ramp audience in crypto.

x402 native

The pay-per-request protocol is built for Base. The payment rail already exists.

16 Technical Architecture



On-chain: vault, staking proxy, fee splitter, ivVVV token, LP pools, auto-compounder, POL engine, burn router. Off-chain: inference router, Venice API, multi-surface interface layer.

17 Open Questions & Next Steps

To Confirm

- Exact VVV staking capacity formula and minimum stake
- Venice ToS on proxying/reselling staked inference
- sVVV receipt token status (live vs planned)
- Rate limits per staked VVV at various tiers
- Venice team appetite for partnership on distribution
- Staking yield distribution mechanics (claim vs auto-compound)
- ivVVV burn discount rate (10-20%) and pricing oracle for VVV/inference

To Build

- ERC-4626 vault contract (needs DeFi security expertise)
- Staking proxy that delegates to Venice's contract
- Fee splitter with configurable distribution (25% perf fee)
- Inference router with capacity allocation logic
- POL management contract for auto-reinvestment
- Burn router: ivVVV → VVV return to vault + inference credit
- First interface: x402 endpoint (proven infrastructure)
- Audit by established DeFi security firm

> **The ask:** This concept requires three parties. Venice provides the models, the privacy, and the 19% native yield. A DeFi team provides audited vault contracts and staking mechanics. We provide the vision, the spec, the agent distribution network, and the interface layer. The protocol sits at the intersection of DeFi composability and AI inference demand, and Base is the only chain where all three converge today.